

Day 5 — TNPSC Group 2A Study Material

Part A – General Studies: Everyday Application: Heat, Nuclear Physics, Laser, Electronics & Communications

READING MATERIAL

The last and densest physics clause in Unit I, bundling four more sub-areas. Nuclear physics is tested via 'match the reaction type' style; heat via direct definitional recall; laser via specific historical/technical facts about the first laser; electronics/communications has no confirmed past-paper evidence in either the local question bank or web search, so treat that portion as lower-confidence general-knowledge coverage rather than exam-pattern-matched content.

Q: Real exam (2025): Match the bomb/reactor type to its nuclear reaction type — Hydrogen bomb, Power reactor, Magnetic confinement of plasma, Atom bomb.

A: Hydrogen bomb → Fission-Fusion reaction. Power reactor → Controlled fission reaction. Magnetic confinement of plasma → Controlled fusion reaction. Atom bomb → Uncontrolled nuclear fission reaction. (Pattern to remember: 'bombs' are uncontrolled/explosive, 'reactors and confinement' are controlled; atom bomb is fission-only, hydrogen bomb combines fission AND fusion.)

Q: What is Celsius also called, and what are the two common temperature units?

A: Celsius is also called Centigrade. The two common everyday temperature units are Celsius and Fahrenheit (Kelvin is the scientific/absolute scale, used less in everyday contexts).

Q: What are the 'boiling point' and 'freezing point' of water, in terms of stable temperature?

A: Boiling point: the temperature at which water boils and the temperature stabilizes (stops rising) despite continued heating. Freezing point: the reference point below which temperatures are expressed with a minus sign.

Q: Where was the coldest temperature on Earth recorded, and approximately what was it?

A: In Antarctica, approximately -89°C .

Q: During which laser process is a photon of energy emitted?

A: Induced (stimulated) emission — this is the core mechanism behind how lasers amplify light (LASER = Light Amplification by Stimulated Emission of Radiation).

Q: Who built the first laser (the Ruby laser), and when?

A: Theodore H. Maiman (T. Maiman), in 1960.

Q: What were the rough dimensions of the original ruby laser's crystal rod, and what surrounded it?

A: About 10 cm long and 0.8 cm in diameter, surrounded by a Xenon flash tube (used to optically 'pump' the ruby crystal with light energy).

Q: What is a semiconductor, and why is it important in electronics?

A: A material (like silicon or germanium) with electrical conductivity between that of a conductor and an insulator. Semiconductors are the foundation of nearly all modern electronics (transistors, diodes, integrated circuits/chips) because their conductivity can be precisely controlled.

Q: What everyday device made satellite-based communication possible?

A: Communication satellites in geostationary orbit relay signals (TV, phone, internet, GPS) across long distances by reflecting/retransmitting them between widely separated points on Earth.

MOCK TEST (WITH ANSWERS)

1. Which type of nuclear reaction does an atom bomb rely on?

- A. Controlled fission
- B. Uncontrolled fission ✓**
- C. Controlled fusion
- D. Fission-fusion combined

Explanation: Atom bomb = uncontrolled nuclear fission reaction. [medium, question-bank-2025]

2. A hydrogen bomb relies on which combination of nuclear reactions?

- A. Fission only
- B. Fusion only
- C. Fission and fusion combined ✓**
- D. Neither — it's a chemical reaction

Explanation: Hydrogen bombs combine a fission trigger with a much more powerful fusion reaction. [medium, question-bank-2025]

3. A nuclear power reactor relies on which type of nuclear reaction?

- A. Uncontrolled fission
- B. Controlled fission ✓**
- C. Controlled fusion
- D. Uncontrolled fusion

Explanation: Power reactors use controlled (regulated, sustained) fission to generate steady heat for electricity. [medium, question-bank-2025]

4. Celsius is also known by which other name?

- A. Fahrenheit
- B. Kelvin
- C. Centigrade ✓**
- D. Rankine

Explanation: Celsius = Centigrade. [easy, web-research]

5. The coldest recorded temperature on Earth (around -89°C) was measured on which continent?

- A. Asia
- B. Arctic (Northern Canada)
- C. Antarctica ✓**
- D. Australia

Explanation: Antarctica. [easy, web-research]

6. The acronym LASER stands for Light Amplification by Stimulated Emission of:

- A. Energy
- B. Radiation ✓**
- C. Electrons
- D. Photons

Explanation: Light Amplification by Stimulated Emission of Radiation. [easy, web-research]

7. The first laser (Ruby laser) was built by:

- A. Albert Einstein
- B. T. Maiman ✓**
- C. Ali Javan
- D. Charles Townes

Explanation: Theodore Maiman, in 1960. [medium, web-research]

8. In the original Ruby laser, the crystal rod was surrounded by which type of flash tube?

- A. Neon
- B. Xenon ✓**
- C. Argon
- D. Krypton

Explanation: A Xenon flash tube provided the optical pumping energy. [hard, web-research]

9. Which of these materials is a classic example of a semiconductor used in electronics?

- A. Copper
- B. Silicon ✓**
- C. Rubber
- D. Aluminium

Explanation: Silicon (and germanium) are the classic semiconductor materials. [easy, authored]

Part B – Aptitude & Mental Ability: Ratio and Proportion

READING MATERIAL

Ratio & Proportion is 1 of 10 named Aptitude sub-skills, and real exam evidence shows it's tested through several distinct shapes, not just 'simplify this ratio'. Learn each shape as its own method.

Q: Real exam (2025): A man spends Rs.500 buying 12 tables and chairs. One table costs Rs.50, one chair costs Rs.40. Find the ratio of chairs to tables purchased.

A: Let tables= t , chairs= c . $t+c=12$ and $50t+40c=500$ (i.e. $5t+4c=50$). Substitute $t=12-c$: $5(12-c)+4c=50 \rightarrow 60-5c+4c=50 \rightarrow c=10$, so $t=2$. Ratio chairs:tables = $10:2 = 5:1$.

Q: Method — combining two ratios that share a common term: If $a:b=3:5$ and $b:c=7:8$, find $a:c$.

A: To chain two ratios, scale each so the SHARED term (b) matches in both. Here b is 5 in the first ratio and 7 in the second — $\text{LCM}(5,7)=35$. Scale $a:b=3:5$ by $\times 7 \rightarrow a:b=21:35$. Scale $b:c=7:8$ by $\times 5 \rightarrow b:c=35:40$. Now $a:b:c = 21:35:40$, so $a:c = 21:40$.

Q: Method — continued proportion (three terms in one ratio): If $a/10 = b/15 = c/20$, find $a:b:c$.

A: All three fractions equal the same constant, so $a:b:c$ is simply $10:15:20$, which reduces to $2:3:4$ (dividing by the HCF, 5).

Q: Method — chaining two ratios via a shared term: If $A:B=3:4$ and $B:C=8:9$, find $A:C$.

A: B is 4 in the first ratio and 8 in the second — scale the first ratio so B matches: multiply $A:B=3:4$ by 2 to get $A:B=6:8$. Now both ratios share $B=8$: $A:B:C = 6:8:9$. So $A:C = 6:9 = 2:3$.

Q: Method — mixture/alloy adjustment (add a pure ingredient to shift a ratio): A mixture of 60 L has acid:water = $2:1$. How much water must be added to make the ratio $1:2$?

A: Acid:water = $2:1$ in 60L means acid=40L, water=20L. We're only adding water (acid stays at 40L). We want acid:water = $1:2$, i.e. water = $2 \times \text{acid} = 80\text{L}$. Water to add = $80-20 = 60\text{L}$.

Q: Method — simplest form of an age ratio: Azhagan is 50 years old, his son is 10. What is the simplest ratio of their ages?

A: $50:10$ reduces (divide both by 10) to $5:1$.

Q: Method — finding an unknown in a proportion (cross-multiplication): If the numbers 2, 5, x , 20 (in that order) are in proportion, find x .

A: In proportion, $2:5 = x:20$, so $2 \times 20 = 5 \times x \rightarrow 40 = 5x \rightarrow x = 8$. (General rule: in $a:b=c:d$, $a \times d = b \times c$.)

Q: Method — the '4th proportional': Find the 4th proportional to 4, 16, and 7.

A: The 4th proportional x satisfies $4:16 = 7:x \rightarrow 4x = 16 \times 7 = 112 \rightarrow x = 28$.

Q: Method — splitting a total using a 3-part ratio and a given average: Three numbers are in ratio $3:5:7$ and their average is 60. Find the smallest number.

A: Average of 60 over 3 numbers means their total = 180. Split 180 in ratio $3:5:7$ (15 parts total): each part = $180/15 = 12$. Smallest = $3 \times 12 = 36$.

Q: Method — 'duplicate ratio' (squaring a ratio): What is the duplicate ratio of $2:7$?

A: The duplicate ratio of $a:b$ is $a^2:b^2$. So the duplicate ratio of $2:7$ is $4:49$.

MOCK TEST (WITH ANSWERS)

1. A woman spends Rs.620 buying 14 pens and notebooks. One pen costs Rs.30, one notebook costs Rs.50. Find the ratio of notebooks to pens purchased.

- A. 2:5
- B. 5:2 ✓**
- C. 3:4
- D. 4:3

Explanation: $p+n=14$; $30p+50n=620 \rightarrow 3p+5n=62$. Substitute $p=14-n$: $3(14-n)+5n=62 \rightarrow 42+2n=62 \rightarrow n=10$, $p=4$. Ratio notebooks:pens = $10:4 = 5:2$. [hard, authored]

2. If $x/8 = y/12 = z/20$, find $x:y:z$ in lowest terms.

- A. 2:3:5 ✓**
- B. 8:12:20
- C. 4:6:10
- D. 1:2:3

Explanation: $x:y:z = 8:12:20$, which reduces (divide by HCF=4) to $2:3:5$. [easy, authored]

3. If $P:Q = 2:3$ and $Q:R = 9:10$, find $P:R$.

A. 3:5 ✓

B. 2:5

C. 3:10

D. 9:20

Explanation: Scale $P:Q=2:3$ by 3 to get $P:Q=6:9$, matching $Q=9$ from $Q:R=9:10$. So $P:Q:R=6:9:10$, giving $P:R=6:10=3:5$. [medium, authored]

4. A 40 L mixture has milk:water = 3:1. How much water must be added to change the ratio to 3:2?

A. 5 L

B. 10 L ✓

C. 8 L

D. 6 L

Explanation: Milk:water=3:1 in 40L means milk=30L, water=10L. Milk stays fixed; we want water= $(2/3) \times \text{milk} = (2/3) \times 30 = 20$ L. Water to add = $20 - 10 = 10$ L. [hard, authored]

5. If 5, 8, x, 24 are in proportion, find x.

A. 12.5

B. 15 ✓

C. 16

D. 20

Explanation: $5:8 = x:24 \rightarrow 5 \times 24 = 8x \rightarrow 120 = 8x \rightarrow x = 15$. [easy, authored]

6. Find the 4th proportional to 3, 12, and 9.

A. 27

B. 36 ✓

C. 30

D. 33

Explanation: $3:12 = 9:x \rightarrow 3x = 12 \times 9 = 108 \rightarrow x = 36$. [easy, authored]

7. Three numbers are in the ratio 2:3:4 and their average is 45. Find the largest number.

A. 40

B. 50

C. 60 ✓

D. 45

Explanation: Average 45 over 3 numbers means total=135. 9 parts total, each part=15. Largest = $4 \times 15 = 60$. [medium, authored]

8. What is the duplicate ratio of 3:5?

A. 6:10

B. 9:25 ✓

C. 9:10

D. 3:25

Explanation: Duplicate ratio = square each term: $3^2:5^2 = 9:25$. [easy, authored]

9. A sum of Rs.450 is divided among 100 boys and girls, each boy getting Rs.5.10 and each girl Rs.3.60. Find the number of girls.

A. 20

B. 30

C. 40 ✓

D. 50

Explanation: Let girls=g, boys=100-g. $3.6g + 5.1(100-g) = 450 \rightarrow 3.6g + 510 - 5.1g = 450 \rightarrow -1.5g = -60 \rightarrow g = 40$. [hard, authored]

10. A father's age (40) to his daughter's age (8) — what is the simplest ratio?

A. 5:1 ✓

B. 8:1

C. 4:1

D. 10:1

Explanation: 40:8 reduces ($\div 8$) to 5:1. [easy, authored]

Part C – General English: Poem: "Sea Fever" — John Masefield

READING MATERIAL

A well-known poem about the pull of the sea, from Masefield's 1902 collection Salt-Water Ballads. Real coaching-site evidence shows heavy figure-of-speech testing per line, plus structure (anaphora, rhyme scheme) — both covered below.

Q: What is the structure and central theme of "Sea Fever"?

A: Three stanzas, each opening with the same declaration 'I must go down to the seas again' (anaphora — repetition at the start of successive lines/stanzas), expressing an irresistible compulsion to return to sailing. Stanza 1 focuses on navigation and sensory details of sailing; stanza 2 on the wildness/freedom of the sea; stanza 3 contrasts the adventurous 'vagrant gypsy life' with the peace found after a long voyage.

Q: Identify the figure of speech: "where the wind's like a whetted knife."

A: Simile — directly compares the wind's sharpness to a 'whetted' (sharpened) knife using 'like'.

Q: Identify the figure of speech: "the wheel's kick and the wind's song and the white sail's shaking."

A: Personification — the ship's wheel 'kicks', the wind 'sings', giving human-like actions to non-human/inanimate things.

Q: Identify the figure of speech: "And the flung spray and the blown spume, and the sea-gulls crying."

A: Alliteration — repeated initial sounds ('flung'/'spray', 'blown'/'spume').

Q: Identify the figure of speech: "And quiet sleep and a sweet dream when the long trick's over."

A: Alliteration — repeated 's' sounds ('sleep'/'sweet').

Q: What is the rhyme scheme of the opening stanza, and what word rhymes with 'shaking'?

A: AABB rhyme scheme. 'Shaking' rhymes with 'breaking' (as in 'a tall ship and a star to steer her by ... And the wheel's kick and the wind's song and the white sail's shaking, And a grey mist on the sea's face, and a grey dawn breaking').

Q: What does "the sea's face" personify, and what other phrases use the same device?

A: It gives the sea a human-like 'face'. Other personification-style phrases include 'the wind's song' and 'call of the running tide' — the sea and wind are repeatedly given human qualities throughout the poem.

Q: Which collection did "Sea Fever" first appear in, and when?

A: Salt-Water Ballads, John Masefield's 1902 poetry collection.

MOCK TEST (WITH ANSWERS)

1. "Sea Fever" begins every stanza with the same line, "I must go down to the seas again." This repetition at the start of successive stanzas is called:

- A. Simile
- B. Anaphora ✓**
- C. Onomatopoeia
- D. Oxymoron

Explanation: Anaphora — repetition of a phrase at the start of successive lines/stanzas. [medium, web-research]

2. "Where the wind's like a whetted knife" uses which figure of speech?

- A. Metaphor
- B. Simile ✓**
- C. Personification
- D. Hyperbole

Explanation: Simile — uses 'like' to compare the wind to a sharpened knife. [easy, web-research]

3. "The wheel's kick and the wind's song" is an example of:

- A. Simile
- B. Personification ✓**
- C. Onomatopoeia
- D. Alliteration

Explanation: Personification — the wheel 'kicks' and the wind 'sings'. [medium, web-research]

4. "And the flung spray and the blown spume" demonstrates which figure of speech?

A. Alliteration ✓

B. Simile

C. Metaphor

D. Hyperbole

Explanation: Alliteration — repeated initial consonant sounds. [easy, web-research]

5. The rhyme scheme of the opening stanza of "Sea Fever" is:

A. ABAB

B. ABBA

C. AABB ✓

D. ABCB

Explanation: AABB. [medium, web-research]

6. Which word rhymes with "shaking" in the opening stanza?

A. Star

B. Song

C. Breaking ✓

D. Face

Explanation: 'Shaking' rhymes with 'breaking'. [medium, web-research]

7. "Sea Fever" first appeared in which 1902 collection by John Masefield?

A. Salt-Water Ballads ✓

B. Collected Poems

C. The Everlasting Mercy

D. Reynard the Fox

Explanation: Salt-Water Ballads (1902). [hard, web-research]

8. The phrase "the sea's face" gives the sea a human-like quality. This is an example of:

A. Simile

B. Personification ✓

C. Hyperbole

D. Onomatopoeia

Explanation: Personification. [easy, web-research]